Harvard Business School

9-699-010

Rev. February 22, 1999

Product Development at Dell Computer Corporation

On an October afternoon in 1993, Mark Holliday, new head of Dell's Portable Division, laid down his papers and greeted key members of Dell's portable computer product development team now entering the conference room. The room occupied a corner of the firm's office park building on the outskirts of Austin, Texas.

Holliday had called the meeting in hopes the group could reach a consensus on recommendations for developing a new line of laptop computers. Dell did not currently market any portable computers. Its first line of portables, discontinued just a year earlier, had had technical problems—just the day before, in fact, the company announced a callback of some 17,000 units. Meanwhile, the lack of portables represented a gaping hole in the company's product line. Given the firm's sagging stock performance, the investment community was closely watching each and every move Dell made.

One way that Holliday and the product development team hoped to differentiate the new Dell line of laptops was through battery life. Batteries lasted less than three hours before needing recharging—a considerable inconvenience for executives trying to work on airline flights or on the road. But the new "lithium ion" (LiOn; pronounced "lee-on") technology being developed at Sony, one of Dell's manufacturing partners, promised to drastically extend laptop usage time without recharging. Unfortunately, since Sony was working to resolve some battery charging issues, LiOn technology was not yet qualified to be used in laptops.

A successful new line of portables, especially with cutting-edge LiOn technology, could catapult Dell back into the portable computer market. As a former engineer with prior computer industry experience on both coasts, Holliday realized, however, that premature commitment to an unproven technology could yield another fiasco for Dell in the laptop market. The discussion around this issue was sure to prove lively.

Professor Stefan Thomke, University of Texas Professor Vish V. Krishnan, and Research Associate Ashok Nimgade, M.D., prepared this case as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation. Names and company-confidential data have been disguised.

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The Personal Computer Industry[1]

Although the first digital computer was designed by Charles Babbage in the 1830s, the limitations of contemporary materials and manufacturing capabilities confined his vision primarily to the drawing board. Not until World War II did armies of engineers create factory-sized computers, such as the 50-foot Mark I at Harvard or the aptly named Colossus in London, capable of undertaking simple mathematical operations or breaking enemy codes.[2] Like dinosaurs, these early computers yielded way to more agile successors.

Technological breakthroughs over the decades allowed for dramatic size reductions with concomitant increases in computing power. Through the 1960s and 1970s, only government (especially defense) and big business could tap the power of computers. But as microchips replaced transistors and wiring, and manufacturing technologies improved, it appeared inevitable that the intersecting trends of miniaturization and lowered prices would ultimately place "microcomputers" at the disposal of individuals.

The microcomputer revolution started in the 1970s, with machines catering to hobbyists and "hackers." In July 1974, an electronic magazine promoted a printed circuit board that came with instruction book for simulations. Known as the Mark 8 computer, over a thousand units were sold at $1,000 each, prompting the appearance of rival products. The market for similar machines grew to nearly 100,000 by 1977, thanks to several key developments: improved microprocessors, a standard operating system, increased availability of software, disk drives (which replaced cumbersome tape storage devices), and cheaper memory. Technology could now put more computing power on someone's desk than existed in the huge 50-foot machines of World War II.

Apple Computer and IBM

Through the late 1970s and 1980s, Apple Computer, a California-based firm, successfully commercialized an intuitively easy-to-use interface. Apple's engineers set the technological pace by cramming in as much new technology into products as possible, often working around the clock for days on end before a market launch. The firm drew on an entirely new cadre of users, primarily in the educational and hobbyist market. Business uses for the small computers, however, remained limited.

Initially, only a handful of larger firms like Texas Instruments and Zenith entered the business segment of the microcomputer market. To put matters in perspective, the Goliath-sized IBM's annual $1.5 billion R&D budget in the early 1980s alone loomed 50% larger than the entire microcomputer market. But IBM, and other giants such as Hewlett-Packard and DEC, which made business-oriented, closet-sized "mainframe" and "miniframe" computers, could no longer ignore the meteoric 30% annual growth of the microcomputer market.

To play catch-up, IBM rapidly leveraged off its traditional corporate base and strong direct sales and service organization. It outsourced hardware and software components to launch its IBM Personal Computer in 1981. Working with Microsoft, IBM created a new operating system that it made available to all personal computer manufacturers; this "open architecture" policy encouraged third parties to develop IBM-compatible software. Apple, in the meantime, kept its technologically superior system proprietary. In a world of two incompatible operating systems, however, the IBM

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[1]Some of the information in this section derives from D. Narayandas, D, and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case No. 596-058.

[2] *Encyclopedia Britannica*, 1991, Chicago: University of Chicago.

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juggernaut eroded Apple market share. By 1983, IBM enjoyed 42% of the microcomputer market, with gross profits as high as 25%.

Enter the Clones

Even IBM could not meet the corporate demand for microcomputers that it had unleashed. During the 1980s, personal computer sales were to grow from ground level to $40 billion. Much of this growth came from manufacturers of IBM-compatible machines, termed "IBM clones." Most notable of these cloners was Texas-based Compaq, which in just five years joined IBM and Apple in the pantheon of the top three-selling retail brand names. Lacking a sales force, Compaq sold through recruited independent full-service retail dealers. (Some 5,000 computer stores had sprung up in the United States by the late 1980s, spurred by an explosion of hardware and software.)

By 1990, microcomputers accounted for 40% of all computers sold, and technological breakthroughs and competition helped drive down the costs of manufacturing the machines. For instance, the cost of processing a million instructions per minute fell from $75,000 in 1980 to $2,000 by 1991. The cost of storing a megabyte of information dropped from $250 in 1980 to $75 in 1991. Lower costs, increasing numbers of software applications, friendlier interfaces (including Microsoft's popular Windows operating system for IBM-compatible computers), and the entry of new manufacturers and distributors all boosted public demand. In 1993, personal computers sold faster than video cassette recorders and almost as fast as TVs.[3] The British journal *Economist* proclaimed that: "Once an icon of technological wizardry, personal computers have become a commodity. . . And customers are less willing to pay for service and hand-holding."[4]

The History of Dell[5]

In 1983, Michael Dell, a freshman at the University of Texas, Austin, started upgrading IBM-compatible personal computers and literally hawked them door-to-door to local businesses. With inventory piling up in his dorm room and crowding his roommate, Dell moved off-campus—and informed his parents of plans to drop out of school at the end of the year. To avoid parental fury, he agreed to return to his pre-med curriculum if business proved disappointing. But with $180,000 in personal computer sales during his very first month, Dell never returned.

He soon started buying and assembling components himself in order to sell computers under his own name directly to customers.[6] High growth rates and attractive margins allowed him to fund growth internally, and he began fielding orders for up to a hundred computers at a time from large oil companies and government agencies. Buyers wanted to meet directly with Dell, however, which created a dilemma: "We had to clean up our workshop," Dell later confessed, "buy some suits and ties, and get ready for meeting America's largest corporations face-to-face."

Dell's seat-of-the-pants operations matched the needs of his computer-literate customers, who demanded quality at a reasonable price. In the years to come, Dell would start a 24-hour complaint hotline and offer a supply of backup replacement equipment. By upgrading IBM-

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[3] S. Lohr, "How Did Dell Computer Stumble?" *The New York Times,* May 28, 1993, Sec. D, p. 1.

[4] *The Economist*, November 2, 1991.

[5] Some of the information in this section derives from D. Narayandas, and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case No. 596-058.

[6] A. Server, "Michael Dell Turns the PC World Inside Out," *Fortune,* September 8, 1997, pp. 76-80+.

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compatibles with the latest microprocessors and peripheral technologies at low costs, Dell's company grew to $6 million by 1985. That year, the firm introduced its own brand of personal computers and ended with $70 million in sales.

By 1990, Dell Computers fielded a broad line of personal computers, winning several trade magazine awards for products and services. Michael Dell himself gained the distinction of becoming the wealthiest person in Texas.

The Dell Business Model

Dell Computer used the same principle to sell computers as Michael Dell had devised during his college venture: eliminate the middle man. The company focused on selling customized products directly via mail to savvy customers. Large clients, which included *Fortune 500* firms, government agencies, and universities, generally demanded reliability and compatibility with existing computers. Competitors in this arena included re-sellers of leading brands such as IBM, Compaq, and Hewlett-Packard. For clients from small to medium businesses, Dell's primary competitors were mail order firms such as Gateway 2000.

Dell serviced its customers with combinations of home-based telephone representatives and field-based representatives. By tracking historical sales records on line, Dell could assess purchase patterns to better respond to customer needs. With direct marketing, in Michael Dell's words, "you actually get to have a relationship with the customer. And that creates valuable information, which, in turn, allows us to leverage our relationships with both suppliers and customers."[7]

Dell assured product quality by extensively pre-testing all the configuration options it offered. Once customers called in orders, customized configuration details were sent on "spec sheets" to the appropriate assembly line (an assembly line in Austin for US customers, and a plant in Ireland for European customers). Assembly proceeded in a manner similar to that for automobiles. It started with a chassis upon which an appropriate motherboard, memory, microprocessor, video card, and other peripherals were installed and then wired. Next, the appropriate software was loaded, and the entire machine was sent to the "burn in" area for four to eight hours of testing. Quality checking occurred at several points along the line. Finally, the machine would be packaged along with manuals and shipped, typically in three to five days after order receipt (a week longer for orders over 100 machines).

A 24-hour telephone support system comprising well-trained technical representatives provided the first post-shipment level of support. This team was able to help Dell's computer-savvy customers diagnose and solve problems themselves 91% of the time.[8] A room in Austin the size of a football field, with 760 cubbyholes with telephones, housed this team within the heart of Dell's sales representatives.[9]

Dell maintained a month's worth of component inventory, but its suppliers generally carried supplemental buffer stock that could be immediately shipped. Outsourcing provided several advantages, according to Michael Dell: "There are fewer things to manage, fewer things to go wrong.

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[7] J. Magretta, "The Power of Virtual Integration: An Interview with Dell Computer's Michael Dell," *Harvard Business Review*, March-April 1998, pp. 73-84.

[8] *Business Week*, July 1, 1991.

[9] S. Lohr, "How Did Dell Computer Stumble?" *The New York Times*, May 28, 1993, p. D1.

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You don't have the drag effect of taking 50,000 people with you."[10] Smaller inventory also allowed new technologies to be adopted quicker, and product development cycles to move faster.

By creating close, coordinated relationships with its suppliers, vendors, and third-party maintenance providers, Dell created the illusion that its customers were dealing with just one large, well-run company. "The supplier effectively becomes our partner," according to Michael Dell.[11] "The rule we follow is to have as few partners as possible. And they will last as long as they maintain their leadership in technology and quality." This relationship also provided Dell with a front-row seat to new technologies its suppliers developed.

Dell Computer Stumbles

By 1990, Dell's success had spawned imitators such as Gateway 2000 and CompuAdd, which also enjoyed low overhead through mail order sales. IBM and Compaq also announced plans for entering the direct mail order business. Compaq, at the same time, sparked off a price war by announcing a 30% cut in its prices.

Meanwhile, feeling the pinch of competition, Dell had expanded into the retail market to attract smaller customers who preferred shopping in showrooms. The new markets spurred Dell's annual sales from $890 million in 1991 to more than $2 billion a year later, overshooting the company's 1992 sales target by half a billion dollars.

The rapid pace caught Dell in a cash crunch. In 1993, after 14 consecutive quarters with rising profits, the firm reported that its profits had been slashed to $10 million, half of what it had projected. Dell stock plunged $7 a share, to $25, on the day of this announcement.[12] Industry observers began speculating about the end of the Dell miracle. Michael Dell, they feared, had lost his legendary attention to detail and his ability to take cues from direct contact with customers and engineers. At any rate, several weaknesses had started to show in the company.

The first problem was that retail selling proved contrary to the spartanism of Dell's traditional direct model. Dell lost the advantage of being able to turn over inventory 12 times a year—twice as fast as its competitors—making it increasingly hard to find a premium in this market. Gateway, in the meantime, stuck to the direct model, growing from $275 million sales in 1990 to $914 million in 1992, with a promise of yet more growth to come.

The second problem involved the lack of senior management capable of guiding the firm toward maturity. Michael Dell became aware of this and hired several senior executives from larger firms such as Motorola to handle Dell's expanding operations.[13]

A third problem involved the lack of structure in Dell's product development process, symbolized, as some thought, by Dell's disastrous foray into the portable computer market.

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[10] J. Magretta, "The Power of Virtual Integration: An Interview with Dell Computer's Michael Dell," *Harvard Business Review*, March-April 1998, pp. 73-84.

[11] Ibid.

[12] S. Lohr, S., "How Did Dell Computer Stumble?" *The New York Times*, May 28, 1993, p. D1.

[13] A. Server, "Michael Dell Turns the PC World Inside Out," *Fortune*, September 8, 1997, pp. 76-80+.

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Product Development at Dell

In accordance with its strategy of commoditization, Dell's R&D budget was smaller than most other large computer firms. This also went hand-in-hand with Michael Dell's philosophy of spartanism that permeated the entire organization. A modest one-story converted industrial warehouse with fluorescent lights and cubicles housed Dell's management and product development professionals.

In this setting, product development at Dell in the early nineties had remained an informal process, run by autonomous teams that often centered around experienced developers. While such an approach delivered several successful products, the results were neither consistent nor predictable. The informality of the process meant that risks were not assessed rigorously in making project investment decisions, nor were the project bounds tightly established. Project execution experienced enormous variability depending on the project leader in charge. In many projects, team members did not all share the same vision of the project's objectives. Designed products were frequently "thrown over the wall" to manufacturing, and in some cases quality issues were addressed too late in the development process.

Some experienced managers began advocating a more structured approach. But engineers, used to a free-wheeling approach, decried formality, which they felt had the potential to lock out creativity. But after one computer project had to be canceled following an investment of several million dollars, Dell's senior management realized the need for a more structured approach adapted to the increasingly rapid product development cycles in the computer industry.

With input from industrial and academic consultants, Dell management started in early 1993 to organize product development around "core teams" of development professionals from several different functions. These teams, led by nominated core team leaders, were to take charge of the product's success from start to finish. Thus, through daily contact between different core team members, and through phase reviews, core teams hoped to avoid the pitfalls of the previous era. At phase reviews, typically slated for every three months, development work was to be reviewed, technical and market risks assessed, and funding for the next phase approved. The new process had the following phases (see **Exhibit 4**):

1. *Profile Phase:* The product development team arrived at a product and market definition, resulting in a two to three page "product features guide".

2. *Planning Phase:* Product team members developed a detailed business case for the product, which senior executives subsequently scrutinized for underlying assumptions and financial impact. This ensures that product developers paid enough attention to business issues before investing in the subsequent expensive phases.

3. *Implementation Phase:* The product development team designed, built and tested functional prototypes of the proposed product. This phase also marked the commencement of the development of instruction manuals and service plans. At the end of this phase, orders for long lead time tooling were placed with suppliers, resulting in a major financial commitment to Dell.

4. *Qualification Phase:* The product development team built production prototypes of the proposed product. Prototypes were distributed to key customers to obtain their feedback on potential product improvements. Salesforce training also began during this phase.

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5. *Launch Phase:* The entire customer buying experience, from opening the packed finished product to running various software applications, was exhaustively tested. Production was ramped up and early customer shipments were made toward the end of the launch phase.

6. *Acceptance Phase:* The team collected customer feedback about the product for up to three months after launch. The phase was completed with an acceptance report that compared the results achieved with planned objectives and included a summary of lessons learned in the spirit of continuous improvement.

Through much of 1993, however, informality prevailed while the new process was being put in place, and core teams often approached the new process in spirit only.

The Portable Computer Industry[14]

For several years, technological and pricing breakthroughs promised to free personal computers from the desktop. Although Osborne marketed the first portable computers in 1981, the machines remained only a small market presence throughout the 1980s. Most weighed over 20 pounds and were derogatorily termed "luggables." Many suffered from severe quality problems because manufacturers viewed the computers as "shrunken desktops."

Nevertheless, by the end of the decade, the quality of portability itself gave these machines gross margins that were typically 3% to 5% above desktops. When several leading computer firms entered the portable field, it was predicted that the market would soon soar. Such predictions forced firms like Dell to determine the appropriate product balance among desktops, portables, and servers (microprocessor systems that created local networks between different computers and communication devices).

Portable computer development relied on several technological breakthroughs: flat liquid crystal display screens developed in Japan that took substantially less space than traditional cathode ray tubes; compact hard drive disks that consumed less energy; and improved battery technology, which allowed over an hour of operations before needing to be recharged.

Manufacturing portables involved less hardware customization than desktops. Laptops, however, often enjoyed more feature differentiation than desktops. Overall, higher workmanship and quality went into these machines given the harsher range of conditions they would face in use. Portables underwent a commensurately challenging array of tests including exposure to shock, vibration, drops, and accelerations.

In 1993, portables were classified as laptops if they weighed between 4.5 and 8 pounds, and sub-notebooks if they weighed under 4 pounds.

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[14] Some of the information in this section derives from D. Narayandas and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case No. 596-058.

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Dell's Latitude Development Project

Early Setback in Portable Computers

In 1991, Dell came out with its first line of portable computers. At Dell's assembly line, the chassis with display and motherboard came prepackaged from outside vendors; only the microprocessor, hard disk drive, and memory were added in the assembly line, followed by software. Through software and communications capabilities, it was possible to create significant personalization. And personalization remained Dell's forte.

In 1992, with portables accounting for 17% of Dell's sales, rumors circulated about quality problems, ranging from faulty battery packs to unreliable screens to frequent power failures to broken hinges. Dell's notebooks, based on older 386 microprocessors, were consequently slower than the 486-based machines fielded by rivals. Even Dell's commodity-like prices could not draw sufficient buyers, many of whom wanted machines that would keep them competitive in the business arena.

Early in the 1993, in what was shaping into the company's *annus horribilis,* Dell canceled a new line of laptops under development, since these were deemed too slow and expensive. A subsequent write-off of $20 million in associated expenses sparked a reorganization of the portables division. When questioned by a reporter about the portables, Michael Dell replied, "One plane is late and these guys claim the whole airline is going down the tube. . . . These guys are babies." [15]

Michael Dell had a point, for the desktops division had racked up unit volume increases of 155% over the previous year. By May, however, notebook sales had slipped to just 6% of Dell sales, at a time when the company had envisioned portables to account for 20-25% of sales. Dell's portable division, one Dell official admitted, suffered from "significant underinvestment" and a "shrunken desktop mentality."[16] Rivals, in the meantime, reported strong portable sales: a Compaq notebook with the more advanced 486 microprocessor, for instance, broke a market launch record for all Compaq products.[17]

In a vote of confidence, however, Bill Gates, chairman of Microsoft and himself a Dell desktop user, stated, "Dell is a super-solid company. They'll get on top of the situation."[18] But Dell went on to post a $75.7 million second-quarter loss. Company officials blamed years of explosive growth, which they claimed strained company performance ranging from inventory management to product forecasting.

Unintentionally rubbing salt in the wound, three owners of Dell portables returned their machines during the fall because they had noted smoke or melted spots on the plastic housing. Investigating engineers found a weakness in an electronic circuitry component: under physical stresses such as jiggling the power cord and AC adapter, the component could sometimes crack.[19] In October 1993, Dell recalled 17,000 notebooks. The company's direct relationship with customers, it turned out, had facilitated tracking down the owners.

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[15] S. Lohr, "How Did Dell Computer Stumble?" *The New York Times*, May 28, 1993, p. D1.

[16] S. Lohr, "Dell's Second Stab at Portables," *The New York Times*, February 22, 1994, p. D1.

[17] T. Hayes, "Dell's Profit and Stock Plummet," *The New York Times*, May 26, 1993, p. D1.

[18] S. Lohr, "How Did Dell Computer Stumble?" *The New York Times*, May 28, 1993, p. D1.

[19] K. Jones, "Dell Recalls Thousands of Notebook Computers," *The New York Times*, October 9, 1993, Sec. 1, p. 47.

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Commenting on Dell's woes, industry analyst Steve Ablondi noted, "This industry moves so fast that you can get thrown off the merry-go-round on one side and still get back on the other side. . . . so Dell can come back from the dead in the portable business."[20] When Dell announced plans for launching a new line of notebooks in 1993, named the "Latitude" series, Michael Dell himself acknowledged that "it's very, very important for us to get this right."

The Next Generation: Choosing which Features to Emphasize

To prepare for the challenges ahead, Dell reorganized its portable division. In Spring 1993, Mark Holliday, the division's new head, immediately formed a core team that was charged with developing a new portable product to be launched in late Fall 1994. The nucleus of the team included Robert Parker and Harry Longbaugh, electrical engineers, Pearl Taylor, a mechanical engineer, Ann Oakley, an industrial designer, and Bill Cody, a software engineer. Henry McCarty, the product marketer, headed the team, which reported to Holliday. Holliday, in turn, reported directly to Michael Dell (see **Exhibit 5**).

Part of the reason for the reorganization was to speed decision-making in an industry with ever-shortening product development cycles. By 1993, it took Dell 18 months to take an idea to production, of which some 14 months were involved with development. Once specification data were available, the process of tooling took 4-5 months. Product life cycles, by way of comparison, lasted roughly three years. The team expected to spend $10 million on the product development effort that would hopefully resuscitate Dell's portable computer line.

The first task the team faced involved taking stock of the situation. Within Dell there was great controversy about the future of the portables market—about its growth, pricing and marketing strategy, and resources need to implement a successful strategy. Within the team there was some debate about whether to enter the subnotebook market rather than just focus on notebooks.

Which features to emphasize formed a large question. Some features engineers had to think about were not immediately obvious to customers—for instance, safety, microchip specifications, and component supply. While product developers thought along 10 or more dimensions, individual customers only thought along two or three. With this in mind, McCarty, the product marketer, and some of the engineers on the team talked with customers, strategized about incorporating new technologies, and even started lining up suppliers.[21]

In the summer of 1993, early in the "Profile Phase," a discussion panel comprising all team members debated which features to highlight in a new line of laptops. Market research indicated that several features rated high in the minds of laptop consumers: price, microprocessor choice, battery life, screen resolution, reliability, weight, and size (see **Exhibit 6**). Dell considered emphasizing battery life—a feature that ranked only behind price and microprocessor choice.

Battery Life as a Value-Adding Feature

The consideration to highlight battery life grew out of a serendipitous encounter in January 1993 in Tokyo. Michael Dell recounted how after a meeting with one of Dell's primary manufacturing partners, Sony, someone ran up to him saying:

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[20] S. Lohr, "Dell's Second Stab at Portables," *The New York Times*, February 22, 1994, p. D1..

[21] Ibid.

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> "Oh, Mr. Dell, please wait one minute. I'm from Sony's power technology company. We have a new power-system technology we want to explain to you." And I remember thinking, Is this guy going to try to sell me a power plant? He starts showing me chart after chart about the performance of lithium ion batteries. "This is wonderful," I tell him. "And if it's true, we're going to put this in every notebook computer we make."[22]

According to a senior manager at Dell, "Michael has a great memory for these types of details, and he filed it away." When the company sought to reenter the laptop market, Michael Dell pulled out this piece of information for his engineers to contemplate.

At the time, most portable computers used nickel metal hydride (NiHi) rechargeable batteries (see **Exhibit 7**). Under optimal conditions, NiHi batteries typically provided less than three hours of power at normal usage. But these batteries suffered from an annoying tendency to recharge only a fraction of their full capacity unless properly managed—a problem referred to as "battery memory." Furthermore, if not properly disposed of, they could release heavy metal toxins into the environment.

In contrast, lithium ion (LiOn), the technology the Sony engineer had mentioned to Michael Dell, suffered from neither of these deficiencies and promised longer recharge lives. LiOn was just starting to be used in camcorders. Thus far, however, no one had figured out how to power larger devices like computers using LiOn technology, especially given the tendency of lithium in larger batteries to separate and even explode. Some battery developers had even given up on the quest, following minor calamities in their laboratories. An unknown number of companies, however, remained in the race for lithium ion technology. Toshiba was known to have a good start, but conceivably Sony engineers might be ahead.

It remained unclear if customers were as excited about new developments in batteries. Focus groups showed that new customers would not necessarily view portables with LiOn as better than portables with traditional batteries. But Michael Dell, did not worry about this. As he pointed out, "The customer doesn't come to you and say, 'Boy, I really like lithium ion batteries. I can't wait to get my hands on some lithium ion.' The customer says, 'I want a notebook computer that lasts the whole day. I don't want it to run out when I'm on the plane.'" [23] Furthermore, among experienced customers—Dell's traditional customer base—battery life proved more important.

Thus, the product development team began viewing the LiOn battery as a tangible feature mirroring the new Latitude computers' overall quality. The new technology could potentially boost the entire product line's market size by anchoring product image in a business with sales cycles of only six to nine months. In the world-wide portable computer market, with annual sales of 11 million units, Dell's management believed that a good Dell product might recapture its old market share of 2.5%, and a superb product might even gain 3%.

The Next Generation: Which Battery?

A new battery technology was not the only computer powering option available to the product development team. Another possibility for doubling battery time could come from allowing users to eject the disk drive and replacing it with a second battery—a strategy that, in Shakespeare's words, "gives to every power a double power." Essentially, Dell could choose between better fuel or a bigger gas tank—two tanks, in fact. The product development team agreed with Henry McCarty,

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[22] J. Magretta, J, "The Power of Virtual Integration: An interview with Dell Computer's Michael Dell," *Harvard Business Review*, March-April 1998, pp. 73-84.
[23] Ibid.

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however, that the dual battery option alone would not make customers perceive the Latitude line as leading technologically.[24]

Thus, at the end of the Profile Phase where a decision had to be made, Holliday and his team entertained only a few realistic development options:

(a) *Continue with a proven battery technology (NiHi):* Dell could continue placing traditional batteries into its Latitude notebooks. McCarty and Taylor, the mechanical engineer, favored this option. This would take less space than the LiOn battery and thus allow for packing in some other feature such as communications control or memory management accessories. All members of Dell's portables division agreed about the viability of this proven option, which would involve no delays and deliver a good Dell product. On the other hand, this option would provide limited product differentiation in an increasingly competitive market.

(b) *Go with the new battery technology (LiOn):* Dell could incorporate new Lithium ion batteries into the notebooks. Parker and Longbaugh, both electrical engineers, championed this option. This option would mean incorporating an unproven and more expensive technology, which would also take up more space than conventional batteries. The new batteries were thicker than traditional batteries, requiring 40 cubic-mm more space—valuable space that could be used for other features. Because of uncertainty over the new technology, other leading manufacturers, such as Compaq, had decided to delay using LiOn technology. Sony (which currently turned out 100,000 rechargeable plain lithium batteries a month for cellular telephones[25]) had yet to complete testing and ramping up production capabilities for LiOn. In fact, if Dell committed to the batteries, it would most likely end up using all of Sony's production for the next few years.

(c) *Defer commitment to either battery technology:* Dell could continue to pursue the laptop development without committing to either battery technology at the current phase review. Buying just two or three months would not pose a major problem, but defering the decision until the qualification phase review, or another nine months—the time McCarty estimated it would take for Sony to provide conclusive test data—was another matter. The Dell team could potentially defer commitment until this point using either one of the following two approaches:

- The team could "over-design" the battery space and the rest of the product to accommodate either battery. This implied that the mechanical engineering and industrial design groups would design the product to ensure that the thicker LiOn battery would fit as well. In addition - and Parker felt this was the harder requirement - the battery charging circuitry and software would have to remain compatible with either system. McCarty felt that the resulting bulkier product would be less attractive to the customer, and the variable cost of over-designing could be as large as 2% of margins on each unit.

- Alternatively the team could follow a "dual path" approach in which two different products are developed independently—one using LiOn and the other using NiHi battery technology. The fixed cost of duplicating engineering activities and tooling meant an additional expense of $2.5 million. These costs stemmed from performing parallel electrical configuration and battery management software, tooling, among other engineering tasks. These estimates, of course, did not include the potential demoralization of engineers that would result from discarding half the craftsmanship in the final product and the opportunity cost of pulling developers away from other projects.

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[24] In addition to addressing the issue of power supply, Dell engineers also were working on power management tools that let users determine power demand. For instance, maneuvers such as dimming the screen or shutting down non-essential programs could lower energy usage.

[25] "Recharged: the Battery Market," *Economist*, May 2, 1992, pp. 323: 82.

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The October Meeting

As he called the meeting to order, Holliday indicated that he would encourage divergent viewpoints on the various options.

Taylor, the mechanical engineer, sparked the discussion: "We really can't afford the risk of delivering one failed product on top of another," she pointed out. "For this round, let's stick to NiHi and get a reliable product out the door. Let's not defer our decision: It runs counter to our new product development process which requires us to commit now. I mean, too much informality and delayed commitment in the past led us to scrap a $20 million product development project midstream. Isn't that why we reorganized the portables division and installed the new development process?"

Parker, the electrical engineer, interjected: "I think we need to get LiOn in our product somehow. That's what will hit our users between the eyes and add value to the entire line. And just think that a successful product could use all of Sony's new batteries and could lock out our competitors from LiOn technology. We could get a strong technical edge on the market."

"That's playing with fire!" McCarty, the marketer and team leader, responded. "I see so many question marks about Sony's supply capabilities that a small LiOn supply could choke our own ability to deliver portables. Remember that we want to use the new technology to build market share? So even if we boost product price to make up for smaller quantity, we may not meet our sales objectives. And let's not ignore the technical risks that make me feel uncomfortable."

"That's why we bother examining all branches of the decision tree," Holliday quietly reminded everyone and thought of the recent analysis that McCarty had prepared for him (see **Exhibit 8**).

"I think deferring our decision is the way to go," said Oakley, the industrial designer. "Worst case, if the LiOn option looks bad, we'd revert to NiHi and lose some valuable internal space that could have supported other features. I know there are other costs, too, but staying flexible isn't a straight gamble; it's just half a gamble."

Longbaugh, the other electrical engineer, threw up his hands. "Listen, you don't rob a stagecoach by perching on the fence. Let's make up our minds one way or the other now. Dell has always had a strong culture of commitment and this doesn't seem right. I, for one, think LiOn is a really neat technology. I suggest we go straight to LiOn, keeping in mind that if it does flop, we could still develop a product with our proven battery technology."

Cody, the software engineer, slipped in the last words before a break in the meeting:

> "True, but if it fails, think of the substantial rework needed—we've estimated the rework to require 70% of the original schedule and 30% of the $10 million development cost. And Henry [McCarty] thinks that the resulting late market entry would likely halve projected unit sales. And any technical fiasco could severely tarnish our reputation, and lead to spillover effects in our desktop business. Let's not fan a brush fire!"

* * * * *

Looking out the conference room windows at the sun setting over the Texan landscape, Holliday reflected on how Dell had helped catalyze Austin's booming high tech economy. Dell had flourished in the Texas Hill Country, relying on its unique model of making and selling computers. Now Wall Street and the business world scrutinized its every move. *The New York Times* had noted,

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"In the fast-paced personal computer industry, the real test of a company's mettle is not if it slips, but whether it can recover quickly. . . . Unless Dell comes back with a strong portable business, the company risks being relegated to second-rate status in the personal computer industry."[26]

Yet after suffering from the prior setback, Holliday personally felt that his product development team had nothing more to lose—not in terms of revenue nor even worker morale. Once again, Holliday watched his colleagues enter the conference room and seat themselves around the table. He then called the meeting back to order. It was time to make a decision.

[26] S. Lohr, "Dell's Second Stab at Portables," *The New York Times*, February 22, 1994, p. D1.

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Exhibit 1 Timeline

1984		Michael Dell founds Dell Computer Corporation.
1985		Dell offers first personal computer of its own design.
1987		International expansion begins, with opening of subsidiary in the United Kingdom. Manufacturing will start there in 1990.
1988		Dell goes public.
1991		Dell introduces its first notebook computer.
1992		The company joins ranks of *Fortune* 500.
1993		
	January	Michael Dell first hears about Sony's lithium ion battery technology in Tokyo.
	March	Amid great uncertainty at Dell about fate of its laptop line, Dell reorganizes its portable division.
	May	After 14 consecutive quarters with rising stock profits, the firm reports that its profits had been slashed to $10 million, half of projected profits. Dell's stock plunged $7 a share to $25 on the day of this announcement.
	August	Market decisions are made about new laptop computer line.
	October	Dell recalls 17,000 discontinued notebooks after three machines were returned by owners reporting such technical problems as smoke or melted spots.

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Exhibit 2 Portable Computer Market Data

(a) Portable Computer Market Size (millions of units)

	World	United States
1992	4.3	2.2
1993	6.2	2.9
1994*	7.4	3.2
1995*	8.9	3.7

* IDC Projections for Laptop Market

(b) Personal Computer Market Size (millions of units) by Vendor

	1987	1989	1991	1992	1993
IBM	28.0	16.9	14.1	11.7	14.0
Compaq	7.5	4.4	4.1	5.7	9.6
Apple	14.0	10.7	13.8	13.2	13.9
Dell	—	0.9	1.6	3.7	5.4
AST/Tandy	2.0	1.7	2.7	2.7	3.6
Gateway	—	0.2	2.5	3.6	4.4
Packard Bell	—	3.3	4.7	5.3	6.7
HP	—	NA	NA	NA	NA
DEC	—	NA	NA	NA	NA
Others	40.0	61.9	56.5	54.1	42.4

Source: *Computer Industry Forecasts and New Games—Strategic Competition in the PC Revolution* by John Steffens (Pergamon Press, 1994). (in D. Narayandas and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case No. 596-058, Rev. March 11, 1996.)

(c) U.S. Personal Computer Market Growth

				Projected	
	1988	1990	1992	1994	1996
Desktops					
Sales (billion $)	20.05	20.78	22.52	25.06	33.0
Units (000s)	8,100	8,750	9,835	11,802	13,100
Portables					
Sales (billion $)	3.28	3.87	4.75	8.48	11.6
Units (000s)	1,130	1,540	2,150	3,800	5,400
Servers					
Sales (billion $)	1.64	2.97	5.47	8.11	12.5
Units (000s)	195	338	457	739	1,250

Source: BIS Strategic Decisions, Inc. (in D. Narayandas and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case No. 596-058.)

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Exhibit 3 Dell Computer Corporation — Selected Sales and Financial Data

(a) Sales by Products and Segments (1991-1993)

	1991	1992	1993
Net Sales ($M)	$890	$2,014	$2,873
Products			
Desktops	90%	88%	94%
Laptops	10%	12%	2%
Servers			4%
Market Segment Sales			
Relationship	59%	61%	64%
Transaction	41%	39%	36%
Markets			
U.S.	72.8%	72.5%	70.9%
Europe	27.2%	27.5%	27.2%
Asia			1.9%

Source: D. Narayandas and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case No. 596-058.

(b) Selected Financial Data (dollars in millions, except per-share data)

	1991	1992
Net Sales	**890**	**2,014**
Cost of goods sold	594	1,545
Gross profit	**296**	**469**
Selling, general and administrative expense	214	310
Depreciation, Depletion & Amortization	14	20
Operating profit	**69**	**139**
Net income (after taxes)	**51**	**102**
Other data:		
EPS (Primary) - excluding extra items and discounted operations	0.18	0.32
Dividends per share	0	0
ROA (%)	9%	11%
ROE (%)	19%	28%
Market value	761	1,705
R&D expenses	42	49

Source: Dell Financial Reports

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Exhibit 4 Dell's New 18-Month Development Process

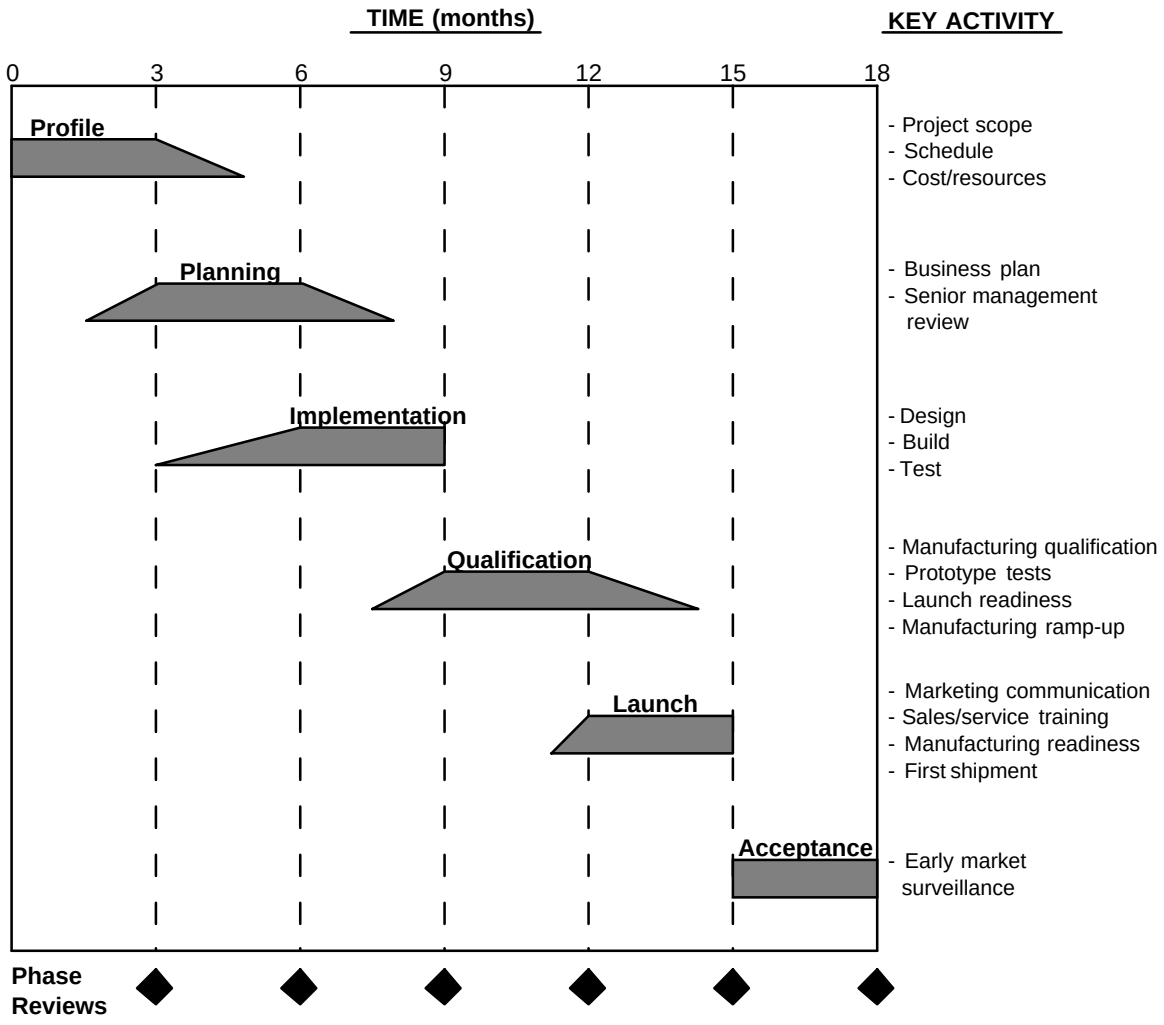

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Exhibit 5 Dell's Organizational Structure and the Notebook Development Team (1993)

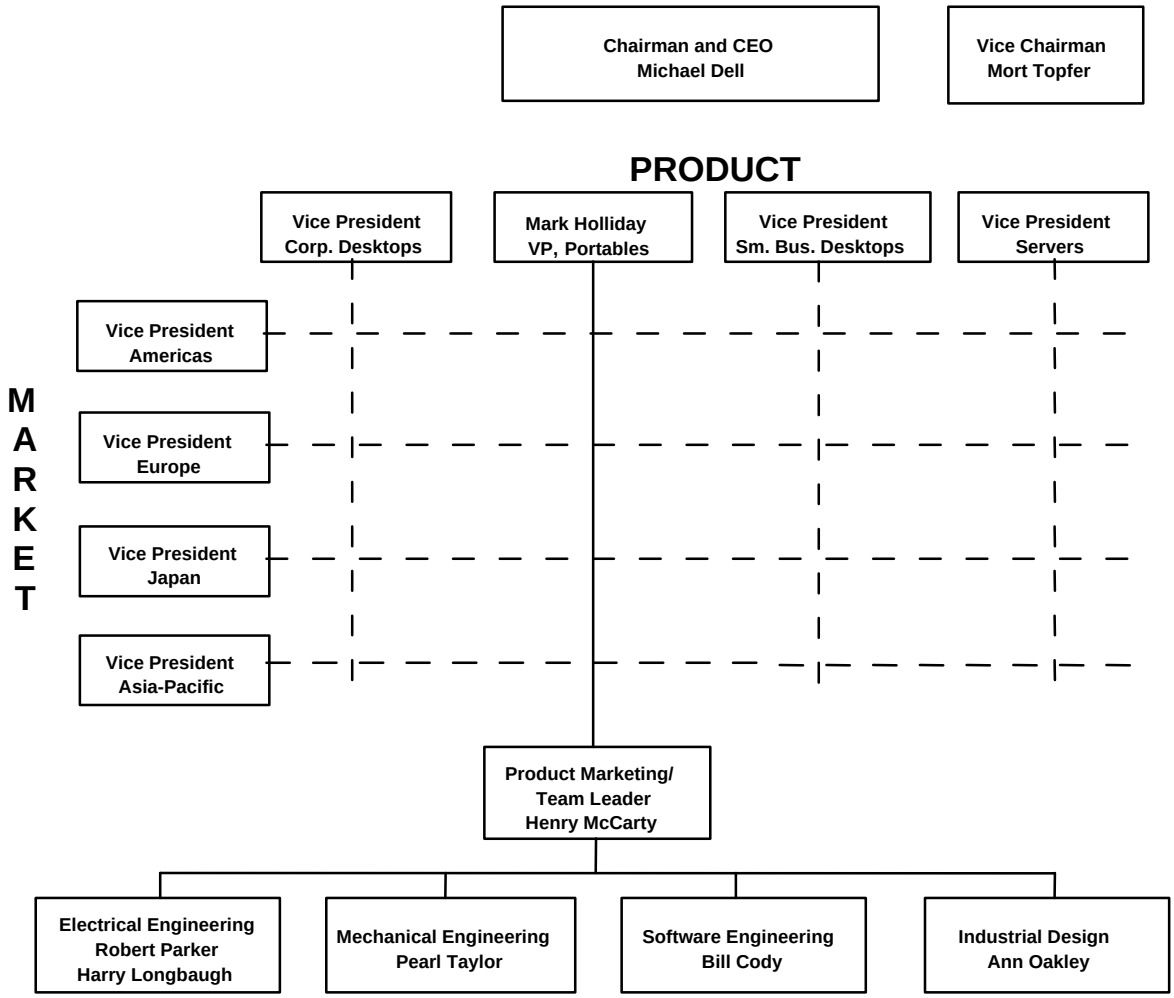

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Exhibit 6 Marketing Memo

MEMO: **PERSONAL COMPUTER MARKET**

ATTENTION: Henry McCarty

FROM: Gordon Lillie, Marketing

DATE: August 15, 1993

Here is the summary market information you requested: we think Dell may capture 2.5% of the market with a good product, but even up to 3% with a superb product. If our gross margins average a fourth of our average price of $2400 over the life of the product, then expected margins over the three-year product cycle would be close to a half-billion dollars. Our models lead us to believe that each month's delay in product development would lead to a loss of about 4% of sales.

Market research shows that most laptop users already have considerable experience with desktops, and that new consumers are more likely to buy a desktop rather than a laptop. But portables are increasingly bought as the primary computer. In 1992, a fourth of portables were bought for this purpose by business customers of all sizes. At the same time, portables were also being purchased to replace desktops. (IDC Customer Directions and Buying Behavior, Year-End Surveys.)

P.S. I am also including marketing information about market size and product features that went into our deliberations.

Top 10 Notebook Evaluation Criteria for Actual Business Customer Purchases

Criterion	Incidence %
Price	54
Microprocessor	35
Battery life	27
Screen resolution	15
Reliability	11
Weight	9
Hard drive storage	6
Durability	3
Compatibility	3
Speed	1

Source: IDC Customer Directions and Buying Behavior, Year-End Surveys.

Top Six Leading Brands for Portables Among 1993 Business Customers

Brand	Purchases
IBM	34%
Compaq	51%
Toshiba	32%
Dell	7%
NEC	8%
Gateway	2%

Source: IDC Customer Directions and Buying Behavior, Year-End Surveys.

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Exhibit 7 Battery Technology

"Life is a search after power."

—Ralph Waldo Emerson

Batteries are energy storage devices that convert chemical energy into electrical energy in order to power attached devices. The first operable battery dates back to 1800, when Italian physicist Alessandro Volta devised the "voltaic pile," a simple arrangement of zinc and silver plates and salt water. Like most batteries to follow, this exploited the different electron-attracting properties of two different metals to deliver sustainable currents. But almost a century of further developments would be needed for the first commercial applications of batteries.

Battery technology remained a slow-moving field through much of the 20th century. Development of rechargeable batteries, however, helped power the automobile revolution, and decades later opened up use of small portable devices such as portable computers. In the early 1990s, computer batteries in widespread use were of the rechargeable nickel metal hydride (NiHi) and nickel-cadmium (NiCad) types. These batteries suffered from two major deficiencies, however: First, they would only recharge up to the point where they had been discharged. Users could take care of this annoying "memory effect" by occasionally allowing fully charged batteries to discharge all the way. Second, the NiCad batteries contained heavy metals that posed environmental health hazards if improperly disposed. (A serious consideration, given that if all the nickel-cadmium batteries produced each year in the Western World were lined up side by side, they would circle the Earth four times.)[a]

These two deficiencies, combined with the short times between recharges, conspired to make battery technology a bottleneck for growth of the portable computer industry. By the early 1990s, manufacturers in Japan (which produced half the world's NiCad batteries), Europe, and the USA raced to develop better products for the $800 million portable computer battery industry.[b] One promising innovation came from lithium ion technology. Plain lithium had already found prior use in the non-rechargeable button-sized batteries that powered watches and cameras for up to years. Chemical innovations now allowed for building lithium ion batteries without size limitations that could be recharged safely. These toxin-free batteries could pack perhaps three times more energy as NiHi battery of the same size and possessed no memory effect. Furthermore, these could be recharged even more rapidly.

But lithium in larger batteries had a worrisome tendency to separate and pose an explosion hazard. Some battery developers had even given up on the quest, following minor calamities in their laboratories. An unknown number of companies, however, remained in the race for lithium ion technology.

A final hurdle for all new battery technologies involved testing. Actually, this was often not much of a hurdle at all given that little standardization existed for testing conditions. Manufacturers, thus, could claim long battery lives based on tests run under low power demand, thus creating a spate of "lies, damn lies, and battery lies." But scrupulous battery manufacturers often hired independent battery testing outfits to conduct two types of tests: "sprint tests," which ran systems at maximal power demand, and "steeplechase" tests, which, mimicked the more typical patterns of intermittent use.

Source: Dell Marketing Division.

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[a] F.E. Katrak, and I.S. Servi, "Battle of the Nickel Links," *Electronic Engineering Times*, February 4, 1991, p. 52.
[b] P.J. Barry, "Valence Technology, Inc.—Company Report," C.J. Lawrence, Inc., 1993.

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Exhibit 8 McCarty's Data on the Different Development Options

Estimates

Forecasted market size (1995-97)	=	11 million units per year
Potential Dell market share (good product, e.g. NiHi)	=	2.5 % (or 825,000 units over 3 years)
Potential Dell market share (superb product, e.g. LiOn)	=	3.0 % (or 990,000 units over 3 years)
Average price per unit over life of product	=	$2400
Average gross margin per unit over life of product	=	$600
Expected product life	=	3 years
Cost of each month delay in development schedule	=	4% of unit sales
Expected length of development schedule	=	18 months (incl. acceptance phase)
Expected cost of development effort	=	$10 million

Option 1: Continue with a proven battery technology (NiHi)

- Confidence = 100% (likelihood that it works as expected)
- Net margin = 825,000 units x $600/unit - $10m = $485 m

Option 2: Go with the new battery technology (LiOn)

- Confidence = 60% (likelihood that it works as expected); risky!!!
- Net margin (if LiOn works) = 990,000 units x $600/unit - $10 m = $584 m
- Net margin (if LiOn fails) = (825,000 units x 0.5) x $600/unit - ($10+0.3x$10) = $234.5 m
- If LiOn fails at launch, a switch to NiHi would require substantial rework (70% of original schedule and 30% of cost). Because competitors would have an established product on the market before them, Dell would lose about 50% of projected units sold.
- If LiOn causes a failure, there could be spillover effects into the desktop business. Dell's reputation for quality could be tarnished.

Option 3: Defer commitment until qualification phase review (dual development or overdesign)

- If dual development paths[a]: Estimated additional fixed cost = $2.5 million
- If product is overdesigned[b]: Estimated additional variable cost = 0.5% of revenue (2% of margin)
- Gross margin (if LiOn works): = 990,000 units x $600/unit = $594 m (before additional cost)
- Gross margin (if LiOn fails): = 825,000 units x $600/unit = $495 m (before additional cost)
- The analysis assumes that Sony will give us enough information at the end of the qualification phase to determine with full certainty if LiOn will work or fail. If it fails, Dell can drop it and revert to option 1.

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[a] These are the actual project costs incurred. They include additional designers and engineers, material and tooling costs, etc. if we follow a dual path until the qualification phase review. The costs do not include the product opportunities we would forego if we had to pull people away from other projects.

[b] Because of the LiOn battery's different dimensions and properties, we would have to "overdesign" the computer case, the charging circuitry and battery management software to accomodate either battery technology which would add about $12 cost per unit.

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